

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Built and shipped to production an autonomous Jira ticket-resolution agent — a multi-step LLM workflow, not a demo: 6-gate triage, MCP-driven context gathering, agentic RAG with FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures — my users sit on the same floor and I prototype against their real questions.
- Built a hybrid orchestrator on top of the analytics agent that lets Product self-serve analytics requests they previously filed as tickets — routes a business question through a co-built semantic-layer skill and the context-engineered agent, reasons about disagreements, and refines definitions over time. Converted 30–60 minutes of analyst execution per covered request into product-led self-service, reducing analyst touch time to near-zero.
- Built a registry-governed semantic layer over BigQuery that integrates LLMs, retrieval, and structured data: one LLM call maps the question to strict JSON intent (the model never writes SQL), a deterministic gate validates and canonicalizes intent against SQLite vocabulary snapshots, and a deterministic compiler emits the SQL — so the same question returns the same answer every time and the one probabilistic step stays regression-testable.
- Built and deployed a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence. Designed a 3-tier event-driven v2 platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Replaced ~20 recurring workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates, turning a recurring 3-hour task into ~3 minutes of setup.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent stack: MCP (Model Context Protocol) server design, sub-agent orchestration, context engineering, prompt engineering, agent orchestration, RAG (FAISS + BM25 + RRF) / retrieval pipelines, evaluation suites (golden sets, human-in-the-loop validation), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

Languages / data: Python, SQL, BigQuery, Databricks, dbt, PySpark, Airflow, MySQL

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Viz: Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL